Data 6 Python Cheat Sheet

This cheat sheet has been modified from the Data 6 Python Reference and includes all of the functions and table methods that you will need for the exams.

Built-In Python Functions

Function	Description	Input	Output
`str(val)`	Converts `val` to a string	A value of any type (`int`, `float`, `NoneType`, etc.)	The value as a string
`int(num)`	Converts `num` to an integer	A numerical value (represented as a `string` or `float`)	The value as an `int`
`float(num)`	Converts `num` to a float	A numerical value (represented as a `string` or `int`)	The value as an `float`
`len(arr)`	Returns the length of `arr`	`array` or `list`	`int`: the length of the array or list
`max(arr)`	Returns the maximum value in `arr`	`array` or `list`	The maximum value of the array
`min(arr)`	Returns the minimum value in `arr`	`array` or `list`	The minimum value of the array
`sum(arr)`	Returns the sum of the values in `arr`	`array` or `list`	`int` or `float`: the sum of the values in the array
`abs(num)`	Returns the absolute value of `num`	`int` or `float`	`int` or `float`
`print(input, ...)`	Prints the input values, separated by spaces	`input`: any inputs to print	`None`
`type(object)`	Returns the type of `object`	`object`: the object whose type is to be determined	`type`: the type of the object

NumPy Array Functions

Function	Description	Input	Output
`make_array(val1, val2, ...)`	Makes a NumPy array with the inputted values	A sequence of values	An `array` with those values
`np.mean(arr)` or `np.average(arr)`	Calculates the average value of `arr`	An `array` of numbers	`float`: The average of the array
`np.sum(arr)`	Returns the sum of the values in `arr`	**array**	`int` or `float`: the sum of the values in the array
`np.prod(arr)`	Returns the product of the values in `arr`	**array**	`int` or `float`: the product of the values in the array

`np.sqrt(num)`	Calculates the square root of `num`	`int` or `float`	`float` : the square root of the number
`np.arange(stop)`, `np.arange(start, stop)`, or `np.arange(start, stop, step)`	Creates an array of sequential numbers starting at `start` , going up in increments of `step` , and going up to but excluding `stop`. Default `start` is 0, default `step` is 1	`int` or `float`	`array`
`np.count_nonzero (arr)`	Returns the number of non-zero (or `True`) elements in an array	An `array` of values	`int`: the number of non-zero values in `arr`
`np.append(arr, item)`	Appends `item` to the end of `arr`. Does not modify the original `arr`	1. `array` to append 2. Item to append (any type)	`array`: a new array with the appended item
`np.cumsum(arr)`	Returns the cumulative sum of the elements in `arr` , where each element is the sum of all preceding elements including itself	`array`	`array`: the cumulative sum of the values in the array
`np.diff(arr)`	Computes differences between consecutive elements in `arr`	`array`	`array`: the differences between consecutive elements in the array containing `len(arr) - 1` elements

String Methods

Function	Description	Input	Output
`str.split(separator, maxsplit)`	Splits `str` into a list of substrings using the specified separator . If separator is not provided, splits at any whitespace. You can also use the optional argument `maxsplit` to limit the number of splits.	1. (Optional) `separator`: the delimiter used to split `str` 2. (Optional) `maxsplit`: maximum number of splits	`list` of substring
`str.join(iterable)`	Concatenates the elements in `iterable` (usually a list or array) into a single string, with each element separated by `str`	`iterable`: an iterable of strings to join (can be an array or list of strings)	`string`: a single string formed by joining the elements of `iterable` with the separator `str`
`str.replace(old, new)`	Returns a copy of the string with all occurrences of the substring `old` replaced by `new`	`old`: the substring to be replaced. `new` : the substring to replace old with.	`string`: a new string where occurrences of old have been replaced by new.

Tables and Table Methods

Function	Description	Input